⚛️ Physics Handbook — Definitions, Formulas & Theories

High-yield Physics for SSC CGL General Awareness / General Science. Each topic: the *definition*, the *formula* (with SI unit), and the *law/theory* behind it.

How to use: learn the law statement + its formula + one real example. SSC asks units, discoverers, and one-line laws most.

🧭 Contents

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📏 Units & Measurement ↑ top

Quantity	SI Unit	Note
Length / Mass / Time	metre / kilogram / second	7 base SI units
Force	newton (N)	1 N = 1 kg·m/s²
Work / Energy	joule (J)	1 J = 1 N·m
Power	watt (W)	1 W = 1 J/s
Pressure	pascal (Pa)	1 Pa = 1 N/m²
Frequency	hertz (Hz)	cycles per second
Electric current / charge	ampere (A) / coulomb (C)	base unit / A·s
Temperature	kelvin (K)	base unit
Electric potential / resistance	volt (V) / ohm (Ω)	—

🏃 Motion & Kinematics ↑ top

Concept	Formula	Definition / note
Speed / Velocity	`v = distance/time`	speed is scalar, velocity is vector
Acceleration	`a = (v−u)/t`	rate of change of velocity
1st eqn of motion	`v = u + at`	u=initial, v=final velocity
2nd eqn of motion	`s = ut + ½at²`	displacement
3rd eqn of motion	`v² = u² + 2as`	no time term
Momentum	`p = m·v`	conserved in collisions

🚀 Newton's Laws & Force ↑ top

Law / concept	Statement / formula
1st Law (Inertia)	A body stays at rest / uniform motion unless acted on by an external force. Defines inertia.
2nd Law	$F = m \cdot a$; force = rate of change of momentum. Gives the unit newton.
3rd Law	Every action has an equal & opposite reaction. (Rocket propulsion, gun recoil.)
Conservation of momentum	Total momentum before = after (no external force). Basis of rockets.
Friction	$f = \mu N$; opposes relative motion. Static > kinetic.
Centripetal force	$F = mv^2/r$; directed to centre (circular motion).

🌍 Gravitation ↑ top

Concept	Formula / fact
Newton's law of gravitation	$F = G \cdot m_1 m_2 / r^2$; $G = 6.67 \times 10^{-11}$ N·m²/kg²
Acceleration due to gravity (g)	≈ $9.8\ m/s^2$; max at poles, min at equator; zero at centre of Earth
Weight	$W = m \cdot g$; mass is constant, weight varies with g
Escape velocity (Earth)	≈ $11.2\ km/s$; $v = \sqrt{2gR}$
Kepler's laws	orbits are ellipses; equal areas in equal time; $T^2 \propto r^3$

⚙️ Work, Energy & Power ↑ top

Concept	Formula	Note
Work	$W = F \cdot s \cdot \cos\theta$	joule; zero if F ⊥ motion
Kinetic energy	$KE = \frac{1}{2}mv^2$	energy of motion
Potential energy	$PE = mgh$	energy of position

Concept	Formula	Note
Power	`P = W/t = F·v`	watt; 1 HP ≈ 746 W
Conservation of energy	—	energy is neither created nor destroyed

💧 Properties of Matter & Fluids ↑ top

Concept	Statement / formula
Density	`ρ = mass/volume`
Archimedes' principle	Upthrust = weight of fluid displaced. (Floating, ships, hydrometer.)
Pascal's law	Pressure applied to an enclosed fluid is transmitted equally. (Hydraulic lift, brakes.)
Bernoulli's principle	Faster fluid → lower pressure. (Aeroplane lift, spin of a ball.)
Surface tension	liquid surface acts like a stretched membrane (droplets, capillary rise)
Viscosity	internal friction of a fluid; honey > water
Hooke's law	`stress ∝ strain` (within elastic limit)

🔥 Heat & Thermodynamics ↑ top

Concept	Fact / formula
Heat (Q)	`Q = m·c·ΔT`; c = specific heat. Unit: joule / calorie.
Temperature scales	Celsius, Fahrenheit, Kelvin; `°F = (9/5)°C + 32`; K = °C + 273.15
Latent heat	heat for phase change at constant temperature (fusion / vaporisation)
Modes of heat transfer	Conduction (solids), Convection (fluids), Radiation (no medium)
1st law of thermodynamics	energy conservation: `ΔU = Q − W`

Concept	Fact / formula
Anomalous expansion of water	water is densest at 4 °C (lets fish survive under ice)

🔊 Sound & Waves ↑ top

Concept	Fact / formula
Wave equation	`v = f·λ` (speed = frequency × wavelength)
Speed of sound (air)	≈ 343 m/s (20 °C); faster in solids > liquids > gases; **cannot travel in vacuum**
Audible range (human)	20 Hz – 20,000 Hz. Below = infrasonic, above = ultrasonic.
Echo	reflected sound; needs ≥ 17 m distance (0.1 s gap)
Doppler effect	change in pitch due to relative motion (siren passing by)
SONAR / ultrasound	uses ultrasonic waves (depth-finding, medical scans)

💡 Light & Optics ↑ top

Concept	Fact / formula
Speed of light	`3 × 10⁸ m/s` in vacuum (fastest known)
Reflection	angle of incidence = angle of reflection
Refraction / Snell's law	bending of light between media; `n = sin i / sin r`
Total internal reflection	basis of optical fibre, mirage, diamond sparkle
Dispersion	white light → 7 colours (VIBGYOR); rainbow
Lens formula	`1/f = 1/v − 1/u`; power P = 1/f (dioptre)
Defects of vision	Myopia (concave lens), Hypermetropia (convex lens)
Scattering	sky is blue, sunset red (Rayleigh scattering — Raman/Tyndall related)

⚡ Electricity & Magnetism ↑ top

Concept	Formula / law
Ohm's law	$V = I \cdot R$ (voltage = current × resistance)
Resistance	$R = \rho L/A$; series adds; parallel = reciprocal sum
Electric power	$P = VI = I^2R = V^2/R$; energy in kWh (unit)
Coulomb's law	$F = k \cdot q_1 q_2 / r^2$ (force between charges)
Faraday's law	changing magnetic field induces EMF (basis of generator, transformer)
Fleming's rules	Left hand = motor; Right hand = generator
AC vs DC	India uses AC, 50 Hz, 220 V domestic supply
Fuse	safety device; melts on excess current (in live wire)

🔬 Modern Physics ↑ top

Concept	Fact
Photoelectric effect	light ejects electrons from metal — **Einstein's Nobel 1921**
$E = mc^2$	mass-energy equivalence (Einstein); basis of nuclear energy
Radioactivity	α, β, γ rays; discovered by Becquerel; studied by Marie Curie
Nuclear fission	heavy nucleus splits (atom bomb, nuclear reactor)
Nuclear fusion	light nuclei combine (the Sun, hydrogen bomb)
Semiconductor	conductivity between conductor & insulator (Si, Ge) — diodes, transistors
Laser	Light Amplification by Stimulated Emission of Radiation

🧠 Key Discoveries & Scientists ↑ top

Discovery / invention	Scientist
Laws of motion, gravitation	Isaac Newton
Theory of relativity, photoelectric effect	Albert Einstein
Raman effect (scattering of light)	C.V. Raman (Nobel 1930)
Electron	J.J. Thomson
Radioactivity	Henri Becquerel; Marie & Pierre Curie
X-rays	Wilhelm Roentgen
Radio / wireless	Marconi (J.C. Bose pioneered microwaves)
Electric bulb	Thomas Edison
Telephone	Alexander Graham Bell
Law of floating bodies	Archimedes

⚡ Quick recall (cover the right) ↑ top

Prompt	Answer
SI unit of force / energy / power	newton / joule / watt
F = ?	m·a (2nd law)
g value	9.8 m/s^2
Escape velocity (Earth)	11.2 km/s
Speed of light / sound	3×10^8 m/s / ~343 m/s
Archimedes' principle	upthrust = weight of fluid displaced
Pascal's law application	hydraulic brakes / lift
Bernoulli	fast fluid → low pressure (aeroplane lift)
Ohm's law	V = IR
Sound cannot travel in	vacuum
Sky is blue because of	scattering of light

Prompt	Answer
Einstein's Nobel was for	photoelectric effect (not relativity)
Water densest at	4 °C
India's domestic supply	AC, 220 V, 50 Hz

Exam tip: SSC Physics favours *units*, *one-line laws* (Archimedes, Pascal, Bernoulli, Ohm, Newton), *discoverers*, and *everyday applications* (why the sky is blue, why ships float). Pair this with NCERT Class 8–10 Science for depth.